these PCBs are labelled as LL+ and LR+ for left and right lamps, respectively. The negative terminals LL- and LR- are connected to resistors R9 and R10, respectively. The sum of current flowing through R9 and R10 is used to charge supercapacitors C1 and C2. Diode D1 prevents discharge of capacitors when PV voltage is not present. For safety, fuse F is included in series with the negative terminals of C1 and C2.

Calculation of current limiting resistors:

Forward drop of yellow LED (Vf) = 2V

Max forward current of LED (If) = 20mA

Total current from one lamp (4-LED array) = 4xIf = 4x20x10^{-3} = 0.08A

Value of current limiting resistance = (18.5 – 2 x Vf)/0.08 = 181.25Ω

Equivalent value of resistances R1 through R4 = 470/4 = 117.5Ω

Value of R9 = 181.25 – 117.5 = 63.75Ω (standard value of 68Ω is chosen)

Please note:

1. Higher value of PV voltage of 18.5V has been considered for the worst case scenario.

2. LED current is maximum when capacitors are fully discharged.

3. Resistors R1 through R8 balance the current flowing through the LED arrays.

Supercapacitor charging circuit

As mentioned earlier, the current flowing through daylamps LL and LR is used to charge capacitors C1 and C2. The charging current calculations are:

Charge required to charge capacitors to 2.5V (Q) = CV = 1000x2.5 = 2500 Coulombs

Total average charging current from R9 and R10 (Ic) = 0.1A (assumed)

Time required for charging = Q/Ic = 2500/0.1 = 25000 seconds or 7 hours (approx.)

Note. The charging current varies from 0.16A to 0.06A, depending on the PV voltage (Vpv) and capacitor voltage, etc. Above calculation assumes an average current of 0.1A to get fair idea about charging time.

It is important to limit the capacitor voltage to less than the rated value. C1 and C2 are rated for 2.7V. Keeping some safety margin, we limit the voltage to 2.5V.

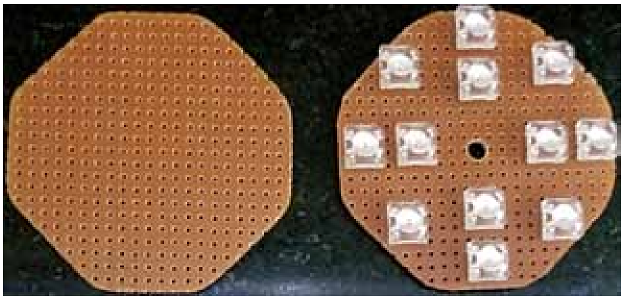

Fig. 3: PCBs for mounting daylamp LEDs (outer 8) and joule thief LEDs (inner 4)

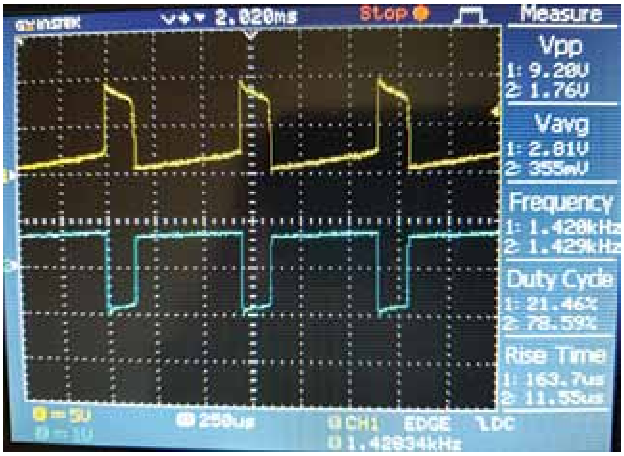

Fig. 4: Joule thief waveform (yellow - collector voltage and blue - base voltage)

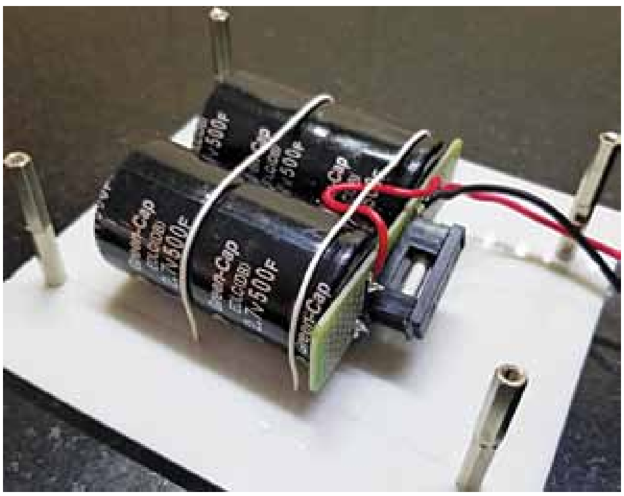

Fig. 5: Supercapacitors along with fuse used in the prototype

At the anode end of diode D1, the circuit comprising diodes D2, D3, and 16 LEDs (LED17 through LED32) is connected. The LEDs are all connected in parallel to form the 'om' sign (a spiritual Devanagari ॐ symbol).

The forward drop of a yellow LED is 2V and forward drop of diode D3 is 0.7V. Therefore, this branch limits the voltage at anode of D1 to 2.7V. Diode D3 has been selected to be of high current rating of 6A. Even though charging current is 0.16A, high current rated diode provides a stable forward drop of 0.7V at all charging current values.

LED17 through LED32 remain off till the capacitor voltage builds up to 2V. After that, these LEDs start turning on slowly. Because, as the capacitor charging current reduces, it gets diverted to these 16 LEDs. When these 16 LEDs are fully on, it means the capacitors have been charged to 2.5V.

In order to ensure that the capacitors do not charge beyond 2.7V, the redundant circuit comprising Zener diode ZD1 and resistor R11 has been provided. Value of R11 has been chosen such that the capacitor voltage does not exceed 2.7V.

Joule thief circuit

The supercapacitors thus charged are used to turn on the joule thief LEDs (LED33 through LED40) once the sun sets. If we directly connect LEDs to the supercapacitor, the LEDs will be on till the capacitor voltage drops to 2V (Vf). After that, the stored energy remains unused.

Joule thief is a popular circuit that is used for extracting the remaining energy from used primary batteries.